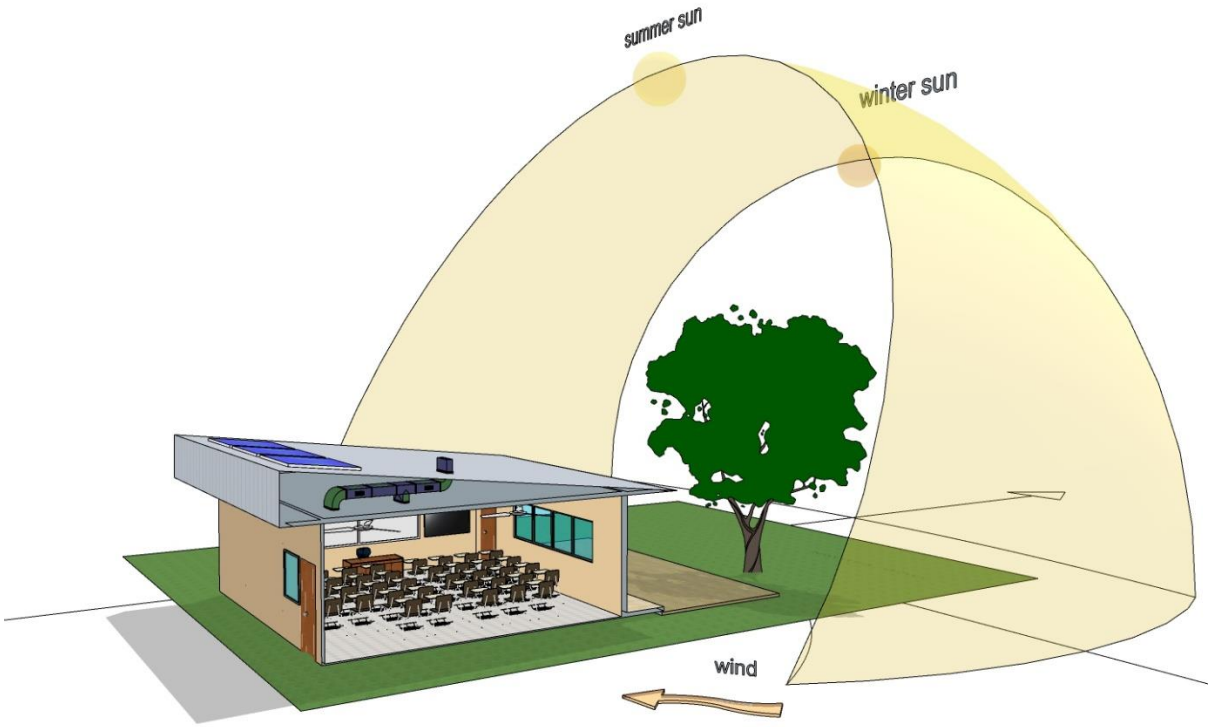

Technical Guideline (TG05X)

Adaptation Measures for Rising Temperature and Heatwaves

Version 1.0 - Draft for Review

October 2025

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Document Control

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Glossary and Abbreviations

Definitions and key terms

Term	Definition
Climate adaptation	The process of adjusting to actual or expected climate and its effects, in order to moderate harm or exploit beneficial opportunities. In building design, this involves integrating strategies that enhance resilience to climate hazards.
Climate change	A long-term alteration in the state of the climate, typically persisting for decades or longer, resulting from natural processes or human activities, particularly the emission of greenhouse gases.
Climate projections	Simulations of future climate conditions based on different scenarios of greenhouse gas emissions, land-use changes, and socio-economic development. These projections are generated using climate models and help inform long-term planning and risk assessments.
Climate resilience	The ability of a system – such as a building or community – to withstand, adapt to, and recover from climate-related shocks and stresses, while maintaining essential functions and thriving in a changing climate.
Extreme weather event	A weather event that is rare for a particular location and time of year. Examples include heatwaves, intense rainfall, cyclones, and bushfires. The definition of "extreme" varies by region and context.
Hazard (climate hazard)	A potentially damaging physical event, trend, or impact – natural or human-induced – that may cause loss of life, injury, property damage, disruption of services, or environmental degradation.
Heatwave	A period of unusually hot and humid or dry weather that lasts for at least two to three days and has a discernible impact on human activities
Representative Concentration Pathways (RCPs)	Scenarios used in the IPCC Fifth Assessment Report (AR5) that describe different greenhouse gas concentration trajectories. Each RCP is defined by its level of radiative forcing (additional heat in the atmosphere) by the end of the 21st century.
Resilience	In the context of this Technical Guide, resilience refers to a building's ability to remain mostly functional during an extreme event and to recover quickly to full functionality.
Scenario	A plausible and coherent description of a possible future climate state, used as the basis for climate projections. Scenarios incorporate assumptions about socio-economic, technological, and policy developments.
Shared Socioeconomic Pathways (SSPs)	The latest IPCC Assessment Report 6 (AR6) identified the following Shared Socioeconomic Pathways (SSPs), SSPs provide baseline narrative scenarios that identify socio-economic and geopolitical assumptions, and economic and technological trends.
Vulnerability	The degree to which a system is susceptible to, or unable to cope with, adverse effects of climate change. Vulnerability includes sensitivity to harm and limitations in adaptive capacity.

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Abbreviations

Abbreviation	Definition
ABCB	Australian Building Codes Board
AIRAH	Australian Institute of Refrigeration, Air Conditioning and Heating
AR	Assessment Report
ASHRAE	American Society of Heating, Refrigerating and Air Conditioning Engineers
BOM	Bureau of Meteorology
CIBSE	Chartered Institution of Building Services Engineers
CCiA	Climate Change in Australia
CMIP	Coupled Model Intercomparison Project
CSIRO	Commonwealth Scientific and Industrial Research Organization
CZ	Climate Zone
DOE	Department of Education
DOF	Department of Finance
DHW	Department of Housing and Works
DSY	Design Summer Year
DTS	Deem to Satisfy
DWER	Department of Water and Environmental Regulation
ESD	Environmentally Sustainable Design
GHG	Green House Gas
GIS	Geographic Information System
HVAC	Heating, Ventilation and Air Conditioning
IL	Importance Level
IPCC	Intergovernmental Panel on Climate Change
ISO	International Organization for Standardization
NCC	National Construction Code
PMV	
RCP	Representative Concentration Pathway
SSP	Shared Socioeconomic Pathway
TG	Technical Guide
TMY	Typical Meteorological Year
UTCI	Universal Thermal Comfort Index
XMY	Extreme Meteorological Year

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Executive Summary

Western Australia's changing climate presents an increasing challenge to the resilience, performance, and safety of government-owned buildings. With rising average temperatures and more frequent and severe heatwaves, the Department of Housing and Works (DHW) recognises the urgent need to integrate climate adaptation measures into the planning, design, construction, and operation of public infrastructure.

This Technical Guideline (TG05X) provides a structured framework to help design professionals, project teams, and asset managers deliver heat-resilient buildings that safeguard occupant wellbeing, maintain essential services, and align with long-term sustainability and emissions reduction goals. It complements the DHW Climate Change Adaptation Guideline (TG050) and focuses specifically on addressing heat-related climate risks across Western Australia's diverse climatic zones.

The guideline aims to embed climate resilience into government building projects through practical design measures, data-driven analysis, and performance-based criteria. It supports a proactive, whole-of-life approach to climate adaptation – ensuring that assets constructed today remain safe, efficient, and functional over their design life through 2050 and beyond.

An internal DHW review identified several gaps in current design and delivery practices, including reliance on outdated or non-future climate data in design models, limited site-specific climate analysis and passive design integration, overdependence on mechanical cooling systems without redundancy, narrow regulatory standards focused on life safety rather than operational continuity, and procurement processes that prioritise upfront cost over lifecycle resilience.

This document outlines a four-stage design process and eight key adaptation strategies to enhance heat resilience in both new builds and retrofits. The framework guides project teams to conduct site-specific climatic analytics using the latest and projected data; apply climate-responsive site planning to reduce heat exposure and improve microclimates; establish thermal comfort targets and cooling strategies aligned with future climate projections; limit external and internal heat gains through passive design and efficient systems; optimise ventilative and mechanical cooling solutions for reliability under extreme heat; and specify heat-resilient materials to ensure long-term durability and occupant safety.

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Implementing these measures will enable WA Government buildings to maintain acceptable thermal comfort during rising temperature extremes and heatwaves, reduce operational disruptions and energy demand through passive and hybrid cooling solutions, support government decarbonisation objectives through reduced greenhouse gas emissions, and protect public health, productivity, and service continuity across critical facilities such as schools, hospitals, and offices.

TG05X establishes a clear and practical pathway for embedding heat resilience into government building projects. By applying evidence-based design principles and incorporating future climate projections, DHW and its partners can ensure that Western Australia's public buildings remain safe, functional, and sustainable in a warming climate. This guideline represents a foundational step toward climate-adaptive infrastructure that supports resilient communities and responsible stewardship of public assets.

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1. Impact of Rising Heat on WA Government Buildings

In Western Australia (WA), the Department of Housing and Works (DHW) oversees the design and operation of thousands of government buildings and houses across the vast land of Western Australia. These buildings are in a variety of climate zones, from hot and humid northern region to temperate regions in the southern, facing different climate challenges and hazards.

The non-residential government buildings - including schools, hospitals, and government offices - serve vital social, cultural, and economic functions. Many are considered critical infrastructure, expected to provide essential services and even shelter during extreme weather events. Understanding which buildings are most at risk, their locations, and the necessary resilience measures is key to ensuring their continued operation.

Climate change and extreme weather events are already impacting these government buildings in a significant manner, causing building failures, damages, service disruptions and have flow-on effects to occupants. This raises concerns from the public and government agencies when projects and assets are seen to be vulnerable to the current and foreseeable future climate hazards (Perkins-Kirkpatrick et al., 2016)

Australia has a long history of designing buildings to withstand extreme temperatures. Traditionally, smaller and older residential structures were built with passive solar design principles – such as effective shading and natural ventilation – at a time when mechanical cooling systems were unavailable. These design features have enabled such buildings to better cope with rising temperatures and heatwaves.

In contrast, many modern buildings – often larger and more reliant on mechanical cooling – lack these passive features. The trend toward expansive windows, reduced eave coverage, and sealed environments has increased dependence on energy-intensive cooling systems, making these buildings more vulnerable to extreme heat events and associated energy demands.

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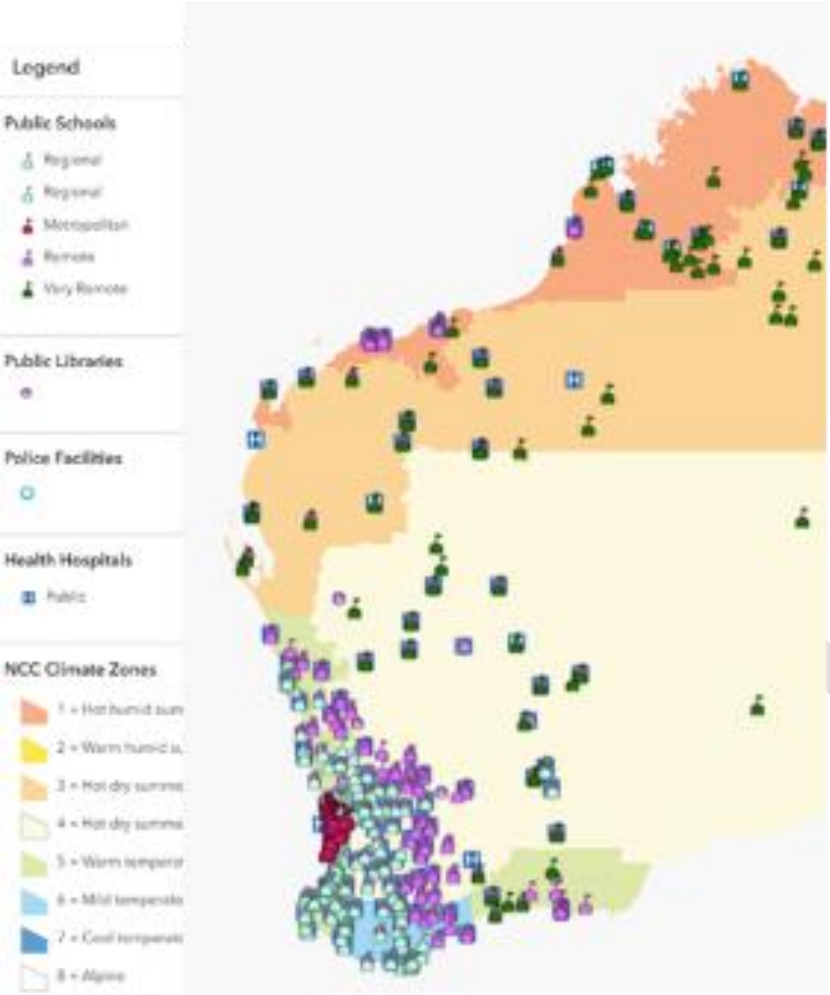

Figure 1 Non-residential Government Buildings in WA

Given that buildings constructed today will likely remain in use for 40 to 60 years, it is essential to consider the long-term impacts of climate change. Projections for 2050, 2070, and 2090 indicate significant shifts in climate conditions (DWER, 2021), which will affect both the structural integrity and operational performance of buildings. However, in practice, today the design and operation of these government buildings are often reliant on building standards and construction codes that are based on historical climate data without consideration of climate change.

Table 1 Key Climate Change Impacts on Australian Buildings

Climate Change Cause	Impact
Chronic impact caused by rising temperature	Increased cooling energy consumption due to higher ambient temperatures
	Reduced energy for space heating and domestic hot water production
	Increased thermal discomfort for occupants
	Structural issues from soil cracking and ground movement affecting foundations and underground services
Acute impact caused by extreme weather events and heatwaves	Increased risk of equipment breakdown and business disruption
	Heat stress and health risks for building occupants in indoor and outdoor spaces
	Elevated bushfire risk

Notable Example:

- In February 2024, WA endured three consecutive heatwaves, breaking temperature records and pushing energy demand to unprecedented levels. This led to the closure of 28 schools in southern WA, affecting over 2,000 students. In Perth, six public schools lost power, leaving students without air conditioning and forcing early dismissals (Hannah Murphy, 2024).

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2. Gaps in Current Building Design and Delivery

An internal review by DHW has revealed several issues in the current design and delivery practices that undermine the resilience of public buildings to rising heat:

- **Outdated climate data:** Many projects are currently using outdated weather files in their building design, such as those in 1980's. Although the use of latest weather data is not mandated in minimum building regulations and codes (e.g. NCC), use of up-to-date climate data is essential to ensuring proper sizing and adequacy of cooling systems. More importantly, there is a lack of consideration of future climate data, and no building projects have been using climate files that have climate change factors incorporated. While it is noted that comprehensive region centric climate files are still being developed, there are data sets from CSIRO available for immediate applications (CSIRO, 2025).
- **Lack of location-specific climate analysis**: Location specific climate analysis is rarely effectively employed for the site planning and building layout design. This has resulted in missed opportunity in utilizing prevailing wind and landscape design to reduce the risk of creating hotspots in buildings.
- **Narrow design standards:** Existing codes prioritise life safety (e.g. structural collapse) but overlook operational continuity during extreme events and thermal comfort under future climate condition.
- **Dependence on mechanical systems:** Many buildings lack passive cooling and shading, making them vulnerable to HVAC failures during heatwaves.
- Single points of failure: Critical services – power, water, communications, sewer – often lack redundancy or robust backup systems.
- **Procurement limitations:** Business cases typically ignore whole-of-life resilience costs and benefits, favouring lowest upfront cost over long-term performance.
- **Weak operational planning:** There is no mandate for monitoring, adaptive triggers, or funding mechanisms to support ongoing upgrades and contingency planning.

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To address these gaps, this guideline introduces a structured approach for design consultants to enhance the heat resilience of building systems in new government projects across Western Australia. It outlines key design steps, relevant resources, and practical examples to support the adoption of best-practice strategies.

While heat resilience must be addressed through integrated interventions across the entire project lifecycle, this guide represents an initial step toward embedding climate resilience into the design process. Future development of complementary guidelines focused on operational controls and adaptive management will be essential to complete the resilience framework.

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3. Heat Resilient Design - Process and Principles

3.1. Design process

Integrating climate responsiveness into building design requires a structured approach that addresses heat risks at multiple levels. The recommended process includes four sequential steps (see Figure 2):

- **Site Planning** – Incorporate local climate projections and neighbourhood context early in the design phase to optimise orientation and landscaping for heat mitigation.
- **External Heat Management** – Apply passive cooling strategies such as solar shading, reflective surfaces, insulation, and thermal mass to reduce heat ingress.
- **Internal Heat Management** – Select low-heat appliances and implement control systems to minimise internal heat loads.
- **Active Cooling Solutions** – Deploy resilient, energy-efficient mechanical systems capable of maintaining comfort during extreme heat events.

Figure 2 Heat Resilience Design Process

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3.2. Key design principles

To reduce vulnerability to rising temperatures and heatwaves, four fundamental strategies should be prioritised:

- **Control solar gain** through shading and glazing solutions.
- **Ensure adequate ventilation** for natural cooling and purge cycles.
- **Provide sufficient insulation** to limit heat transfer.
- **Incorporate thermal mass** where appropriate to stabilise indoor temperatures.

The most effective adaptation measures combine solar shading, enhanced insulation, and natural ventilation. However, their performance may diminish under future climate conditions. For example:

- Natural ventilation becomes less effective as outdoor temperatures rise.
- Increased insulation without purge ventilation can trap heat overnight.
- Air conditioning efficiency declines in extreme heat and can be maladaptive, contributing to urban heat and greenhouse gas emissions if powered by carbon-intensive energy sources.

3.3. Technology and industry considerations

The building industry has access to a wide range of products and technologies to improve heat resilience. To align with emissions reduction goals, preference should be given to passive solar design and low-energy solutions. Active systems should only be used when powered by renewable energy or designed for high efficiency.

Given the building industry's reliance on codes (e.g., NCC), standards (e.g., AS1668.2) ((Standards Australia, 2024)), and rating systems (e.g., Green Star), achieving heat resilience through compliance and certification represents a practical and effective pathway for project teams.

- Planning and development regulations
- Building codes and standards
- Building rating systems
- Individual choices and operational practices

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3.4. Implementation and cost analysis

Adaptive measures should be evaluated for cost-effectiveness and impact based on project-specific needs and site conditions. Detailed design solutions should be tailored to specific sites or projects and assessed for both effectiveness and cost. Project teams can then prioritise measures based on impact and feasibility. For example, installing ceiling fans in large-volume spaces offers a low-cost, high-impact improvement for heat resilience and can be implemented immediately as a quick win.

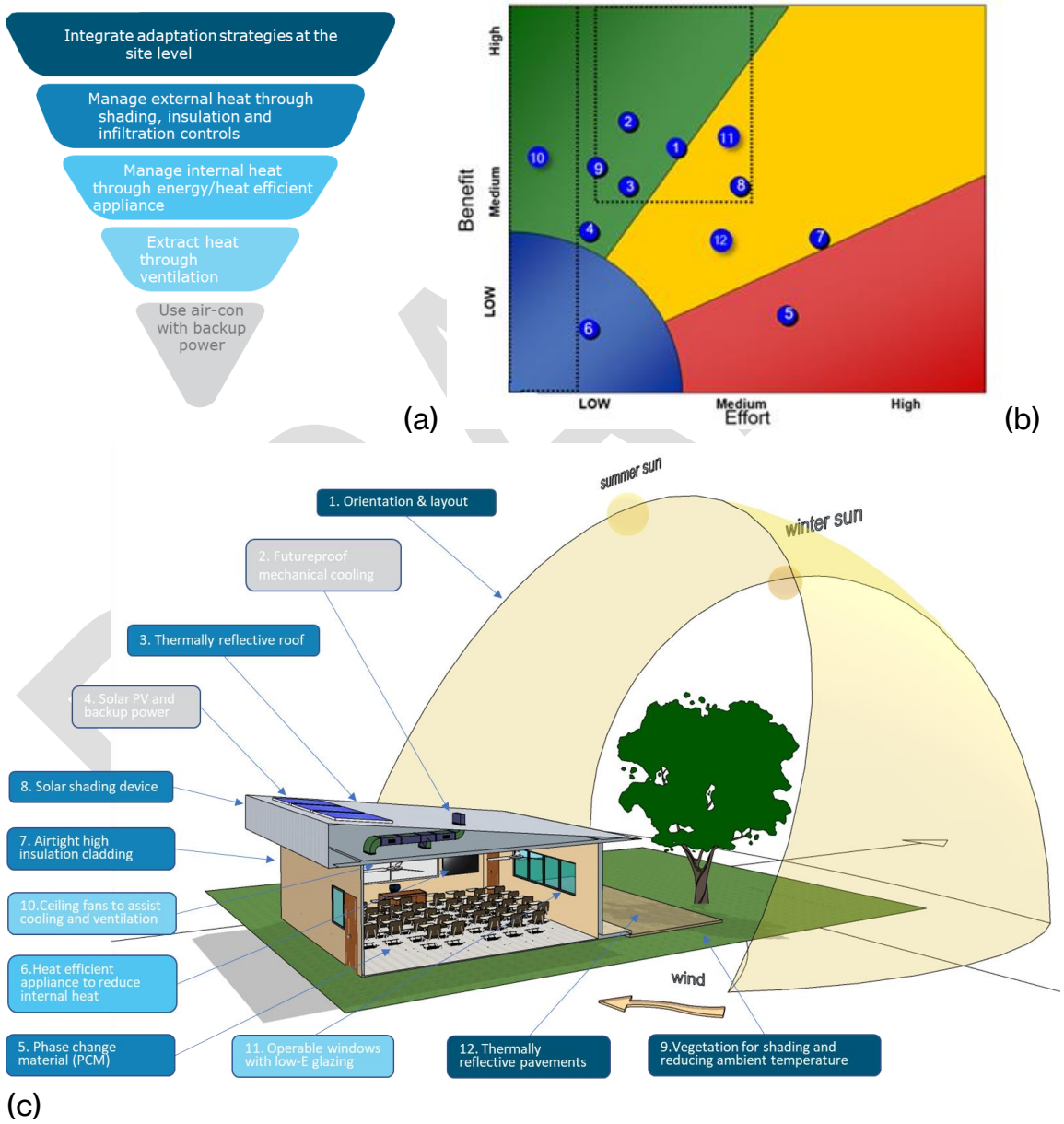

Figure 3 Heat Resilience Design Solutions - Overview & Prioritisation

(a) Design hierarchy; (b) Prioritisation matrix; (c) Overview of adaptation strategies

4. Adaptation Measures for Heat Resilience

Effective heat resilience requires a holistic approach that combines passive design principles, efficient active systems, and context-specific planning. This includes controlling solar gain, enhancing insulation, optimising ventilation, and incorporating thermal mass, supported by renewable-powered cooling solutions where necessary. By embedding these measures into both new construction and retrofits, the built environment can better withstand future climate conditions while aligning with sustainability and emissions reduction goals.

This section presents eight recommended adaptation strategies **aligned with TG050: Climate Change Adaptation Guideline for Non-Residential Government Buildings Heat Resilience Control Measures** (DHW, 2025)(see Appendix D). The following sections illustrate each strategy with examples, aiming to support project teams in effectively applying these measures during implementation.

4.1. Site-specific climatic analytics

The first important strategy that project team should consider is to carry out a site-specific climate analysis and embed climate responsiveness into early stage of the design process to inform the decisions of site planning and cooling target setting.

While climate responsive design is briefly mentioned in DHW and agencies' design requirements, consultants should put them into practice and incorporate them into the design process, particularly in determining building layout and massing.

- **Research and analyse current climate data** - obtain the latest site-specific weather files (after 2015) from reliable sources such as BOM, NATHERS and AIRAH websites (AIRAH, 2022); study the trends of key climate parameters such as temperature and humidity, identify historical heatwave events.
- **Research and analyse future climate data** – obtain future site-specific climate data (including both average and extreme situations) for various projection scenarios (e.g. RCP8.5 and RCP4.5). These climate data may be available at CSIRO and Climate Change in Australia portal or from DWER (CSIRO, 2025). Use Geography Information System (GIS) to ensure only site-specific data is acquired for project design.

- **Conduct climate analysis to form general strategies** - using established software to identify opportunities of passive cooling strategies, e.g. if the site conditions favour the use of natural ventilation, if the humidity level on site would allow effective evaporative cooling.

Notable Example:

- Below example shows a high-level climate analysis of a high school project located in Roebourne, WA (Figure 4). The results show a limited opportunity to use passive design in that project (Figure 5, Figure 6).

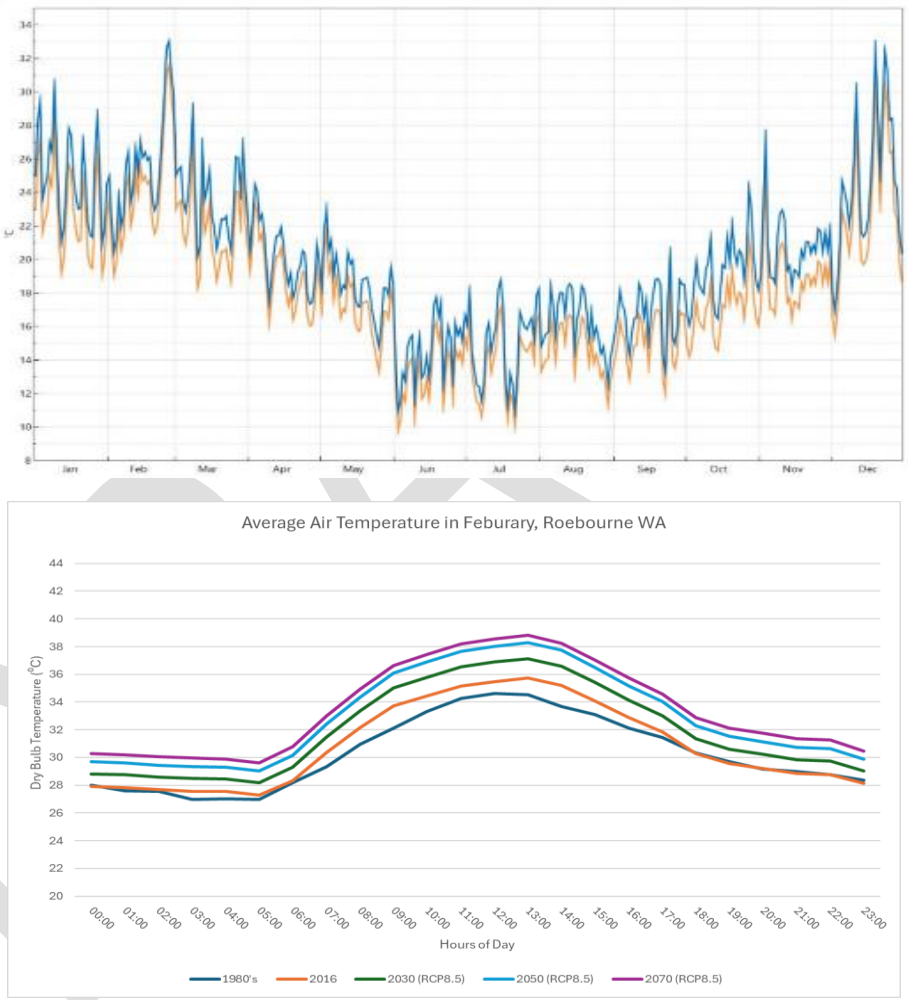

Figure 4 Average Temperature in Roebourne

Top: (Yellow line) Historical temperature based on typical weather year (TMY, 1990-2015); (Blue line) Projected temperatures in 2050 (RCP8.5). (Source: CSIRO)

Bottom: Average temperature in February in multiple time horizons

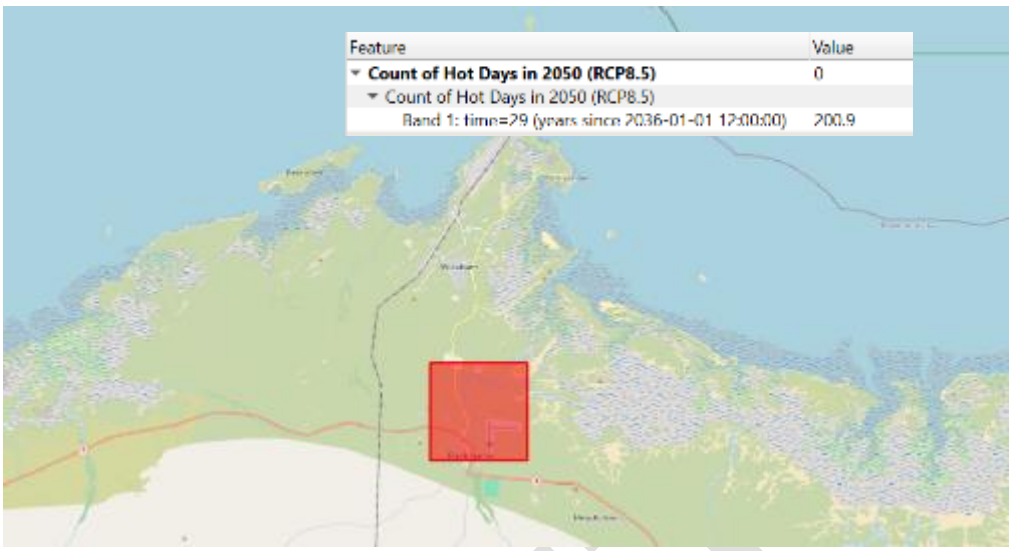

Figure 5 Use of GIS to identify future heatwave risks

GIS is applied to future climate file obtained from CSIRO and BOM (CCiA website) to identify change of climate parameters, indicating more than 200 days per year where the temperature will exceed 35ºC in 2050.

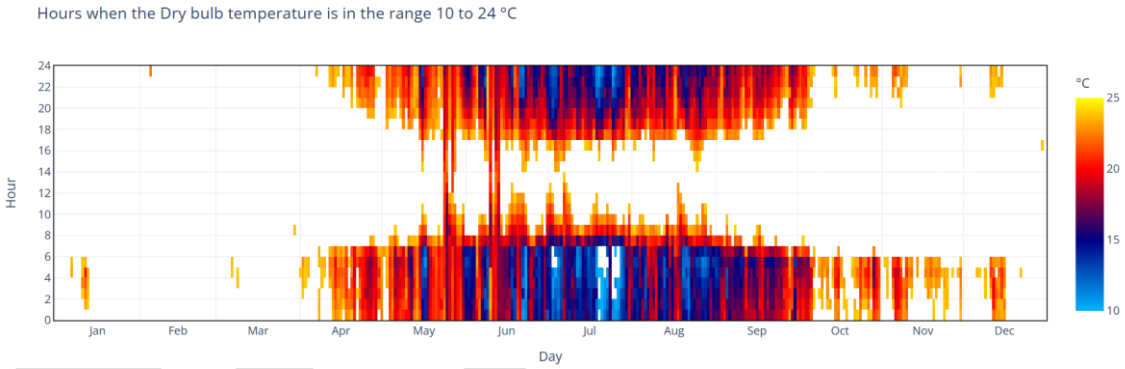

Figure 6 Natural ventilation opportunity for a project in Roebourne

Climate analysis suggests the hours suitable for natural ventilation (between 10ºC to 25ºC) are very limited during school hours throughout the year in the Roebourne. Hence, the cooling strategy for this project should not be based on natural ventilation.

4.2. Climate-responsive site planning

The next step is to explore ways to cool the site and remove potential hot spots in the site planning. This not only provides more conducive outdoor thermal comfort, but also reduce the thermal load for building systems to process due to the reduced need to treat ambient air intake for indoor occupants.

This step entails several strategies such as:

- Use of vegetations to maximise shading effect and use of evapo-transpiration effect of vegetation to reduce heat
- Use of cool materials in soft and hard landscaping design for shading and thermal modulation, etc.
- Placement of “hot spots”, such as carparks, heat-rejection units of air-conditioners, to places with good ventilation and distance away from occupancy areas
- Optimisation of buildings layout, massing to create wind pathways to facilitate airflow to remove the heat from the site.

Notable Examples:

- Site-level microclimate analysis can be a valuable tool in identifying how landscape design can enhance thermal comfort across a campus. Simulations demonstrate that shaded areas created by buildings and vegetation can significantly reduce direct solar radiation, forming cooler microclimates or “cool spots.”
- In a school project in Roebourne, strategic placement of landscaping and vegetation was shown to improve the outdoor thermal environment. Through insulation and the cooling effects of evapotranspiration, these elements help regulate the local microclimate and enhance comfort for occupants.
- As illustrated in Figure 7, extensive greenery has been incorporated to reduce ambient temperatures experienced by students and staff. Preliminary studies indicate that tree shading alone can lower perceived ambient temperatures by approximately 2–3°C, contributing meaningfully to outdoor comfort and wellbeing.

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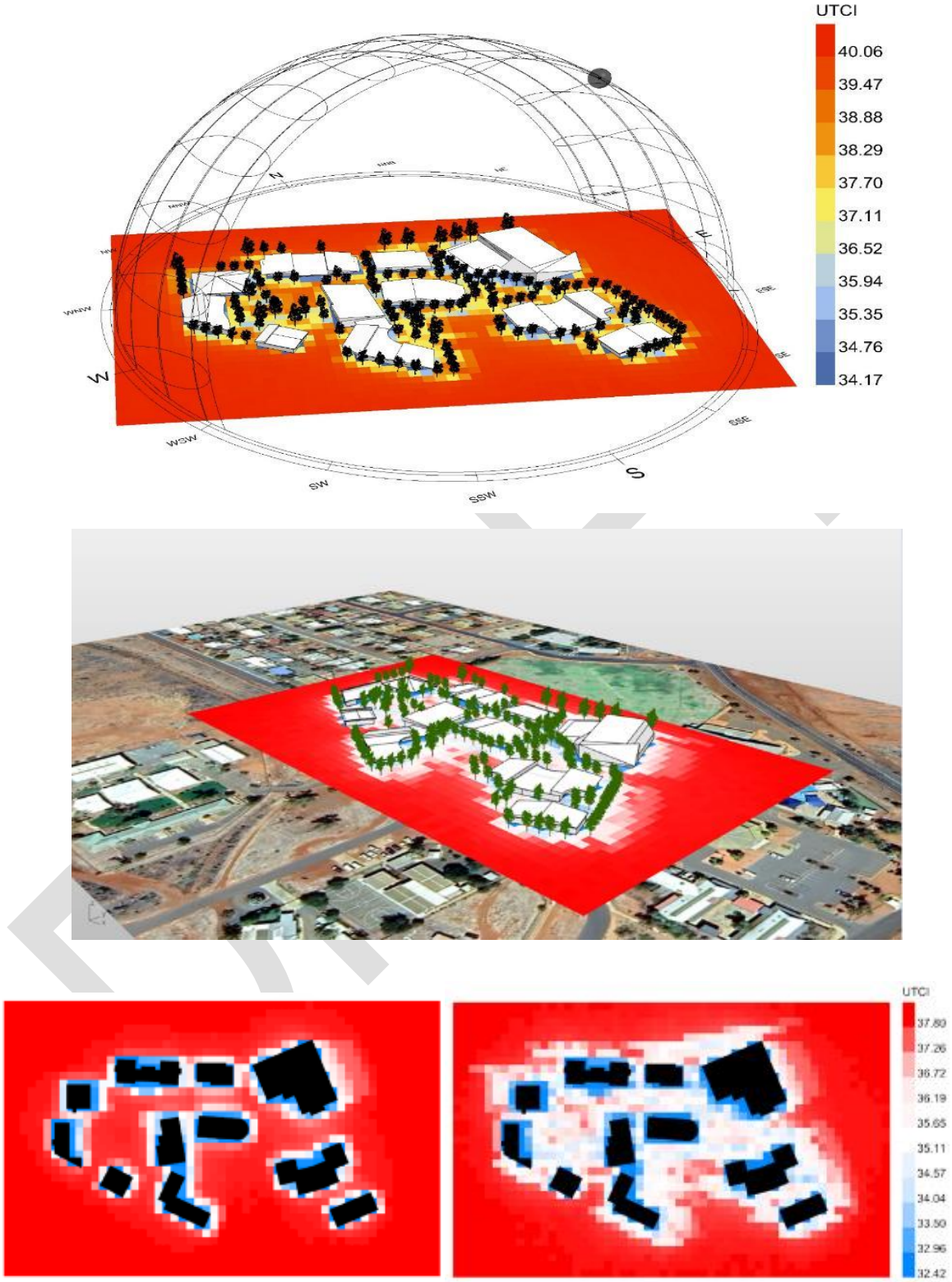

Figure 7 Temperature reduction of 2-3ºC via tree shading at Roebourne DHS

Top & Middle: Campus thermal map during the peak hot day / hour (Feb 8, 3pm)

Bottom: perceived ambient temperature (UTCI) before (left) and after (right) and landscaping

- Beyond solar shading, climate-adaptive design should also consider strategies to enhance the site's wind environment. This includes maximising air movement to support ventilative cooling, particularly during heatwaves. Simulations conducted under extreme heat conditions can offer valuable insights for architects and structural engineers, helping to optimise building geometry and site layout to both improve airflow and withstand extreme wind events (Figure 8).

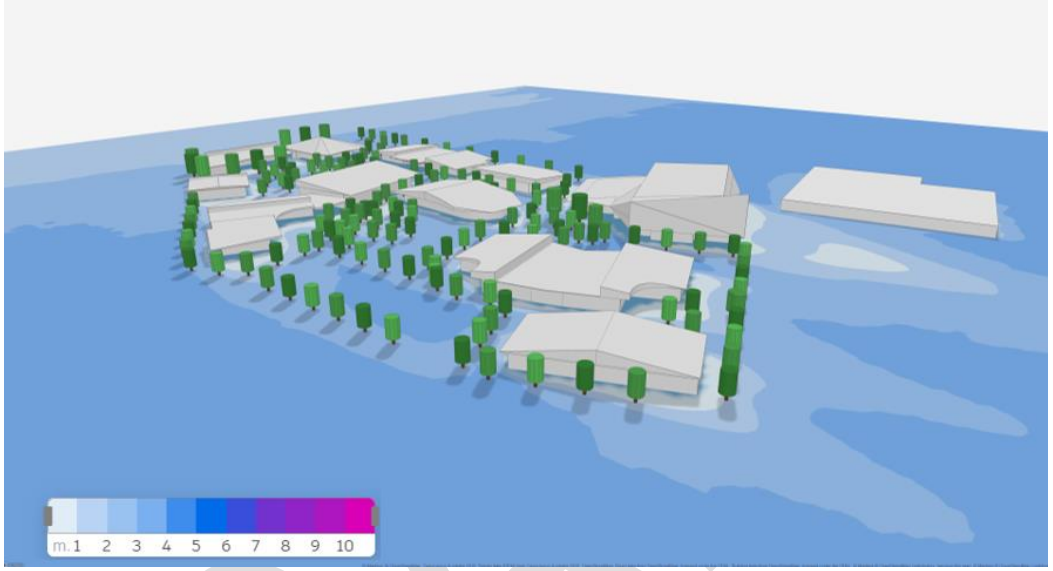

Figure 8 Wind simulation for a school to optimise site layout

4.3. Establishing targets and cooling strategies

Thermal comfort targets should be defined based on both macro and micro-climatic conditions surrounding the building. These targets must reflect the specific function of each space and the needs of its occupants, taking into account asset criticality and occupant vulnerability. Design parameters should address performance under typical operating conditions as well as during extreme events, such as heatwaves. Importantly, they must also consider projected temperature increases over the building's design life due to climate change, ensuring long-term resilience and occupant wellbeing.

Thermal comfort for temperature rise[1]

[1] This section references Clause H1 in Appendix A of the TG050: Climate Change Adaptation Guideline for Non-Residential Government Buildings.

Thermal Comfort (TC) design targets and evaluation criteria should be defined early in the design process, incorporating projected climate change impacts and the specific vulnerabilities of building occupants.

To assess whether thermal comfort design targets are being met, dynamic simulation is essential. These simulations should incorporate projected climate change data to evaluate building performance under future conditions. Project teams are encouraged to use the SSP3-7.0 scenario for the years 2050 or 2070 – or an equivalent high-emissions projection such as RCP8.5 – to ensure the design remains resilient throughout the building's intended lifespan (IPCC, 2013, 2023).

The following recommended criteria provide a framework for setting appropriate targets.

<u>For mechanically ventilated spaces:</u>

- Achieve a Predicted Mean Vote (PMV) between -1.0 and +1.0 for not less than 95% of the occupied floor area for not less than 98% of the building's operational hours, based on suitable future weather files.
- For spaces for vulnerable occupant groups, such as the elderly, disabled, infirm or very young children, achieve a PMV between 0 and 1.0 for 95% of the occupied floor area for 98% of the building's operational hours, based on suitable future weather files.

  [Notes: Refer to NCC 2022 NCC J1V3.1(b); ISO 28803:2012 Ergonomics of the Physical Environment – Application of International Standards to People with Special Requirements]

<u>For building spaces that rely on naturally ventilation or mixed mode ventilation systems:</u>

- Achieve acceptable operative temperatures with at least 80% acceptability across 95% of the occupied floor area for 98% of the building's operational hours following the adaptive thermal comfort model, based on suitable future weather files.

<u>For spaces for vulnerable occupant groups, such as the elderly, disabled, infirm or very young children:</u>

- Achieve acceptable operative temperatures with at least 90% acceptability across 95% of the occupied floor area for 98% of the building's operational hours following the adaptive thermal comfort mode, based on suitable future weather files.

  [Note: Refer to NCC J1V1 Guide, ASHRAE Standard 55-2020 Thermal Environmental Conditions for Human Occupancy.].

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Thermal comfort for heatwaves[2]

Buildings should be designed to maintain acceptable thermal comfort for occupants during heatwaves. While it may not be feasible to ensure comfort in all spaces at all times during extreme conditions, priority should be given to critical areas – particularly those serving vulnerable populations or essential functions.

To evaluate performance under future heat stress, dynamic thermal simulations should incorporate climate change projections. Recommended datasets include the Future Extreme Meteorological Year (fXMY) and Future Design Summer Year (fDSY). Where these are unavailable, CSIRO's projected Typical Meteorological Year (TMY) files – based on the RCP8.5 or SSP3-7.0 scenario for the year 2090 – may be used as proxies to assess long-term resilience (Ren et al., 2022).

The following design targets are recommended:

- For critical spaces serving vulnerable user groups in Importance Level (IL) 3 or IL 4 buildings – such as kindergarten classrooms or patient wards – indoor thermal conditions should comply with the CIBSE TM52 Category I standard, using future weather files for simulation.
- In mechanically cooled spaces, the number of overheating hours – defined as periods exceeding the PMV/PPD or operative temperature thresholds – should not exceed 3% of total occupancy hours during the summer months (December, January, and February).
- In naturally ventilated or mixed mode ventilation spaces, overheating hours – defined as periods exceeding the TM52 criteria – should not exceed 3% of total occupancy hours during the summer months (December, January, and February).

  [Note: Refer to TM52-2013 The Limits of Thermal Comfort: Avoiding Overheating, Section 6.1 and 6.2.]

Notable Example:

- The example below illustrates the projected increase in cooling demand and decline in thermal comfort for a school located in Perth (Climate Zone 5) under both current and future climate conditions. Due to rising temperatures driven by climate change, the school's cooling demand is projected to increase by approximately 22% by 2030 and 61% by 2070 (see Figure 9). The existing HVAC system, designed for present-day conditions, will likely be insufficient to meet future cooling loads. Overheating hours – defined as occupied periods during which indoor

[2] This section references Clause H2 in Appendix A of the TG050: Climate Change Adaptation Guideline for Non-Residential Government Buildings.

operative temperatures exceed 26°C – are projected to surpass the acceptable threshold outlined in CIBSE TM52 (i.e. no more than 3% of annual occupied hours) by 2030. By 2070, which aligns with the end of the school's design life, these overheating hours are expected to more than double, indicating a significant decline in thermal comfort and highlighting the need for proactive design interventions.

- This example highlights the critical importance of incorporating climate change considerations into the design of building thermal systems. Notably, the use of climate change factors based on the RCP8.5 scenario is now officially recommended by the Australian Building Codes Board (ABCB) and reflected in the draft of the National Construction Code (NCC) 2025.

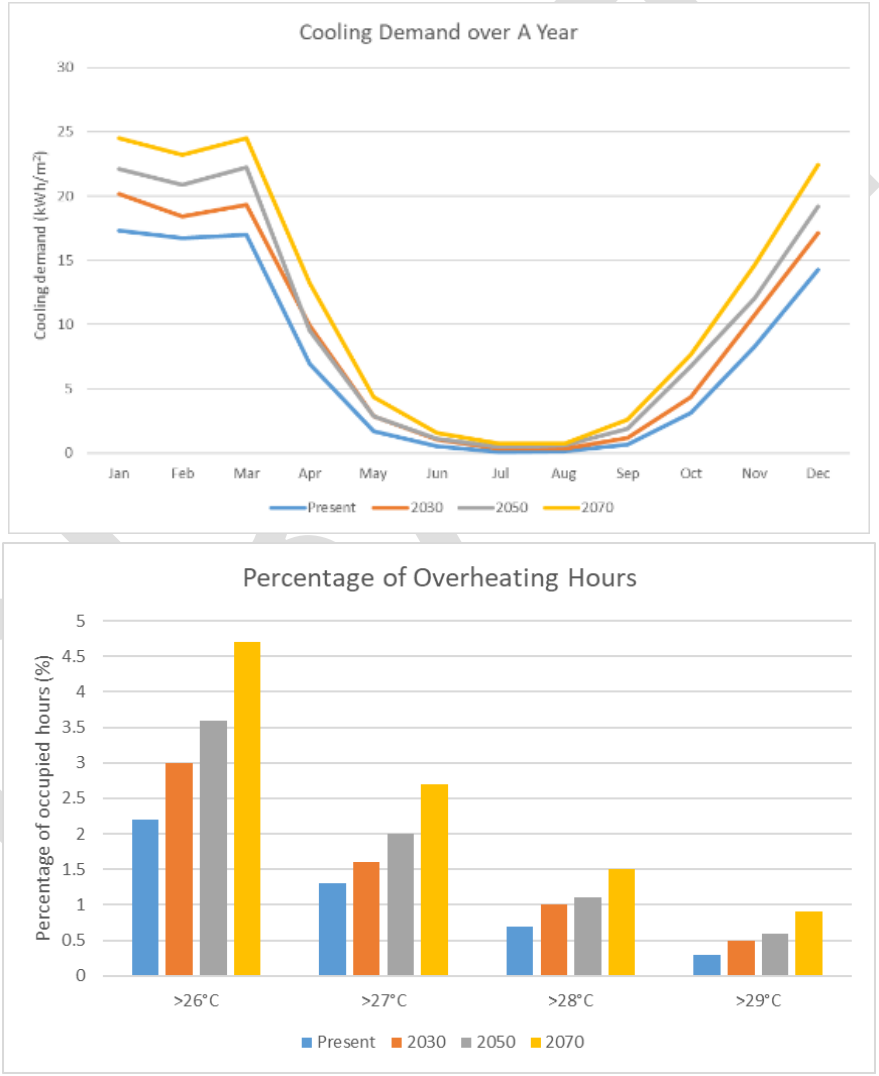

Figure 9 Increase cooling load and overheating hours due to climate change

Determining overall cooling strategies

Once outdoor climate conditions and indoor thermal comfort targets have been established, project teams should develop a comprehensive cooling strategy.

This process should follow the structured approach outlined in the *Northern Regions Heating, Ventilation and Air Conditioning Design Guidelines (DOF, 2017)*, which supports the identification of appropriate cooling modes – mechanical, natural, or mixed – for different building zones (see Figure 10).

To identify cooling strategies that align with local climate conditions and indoor thermal comfort requirements, computational psychrometric analysis can be applied. This technique enables the assessment of various passive and hybrid cooling options, including evaporative cooling, solar shading, and the use of thermal mass (see Figure 11), helping to optimise comfort and energy performance under projected climate conditions.

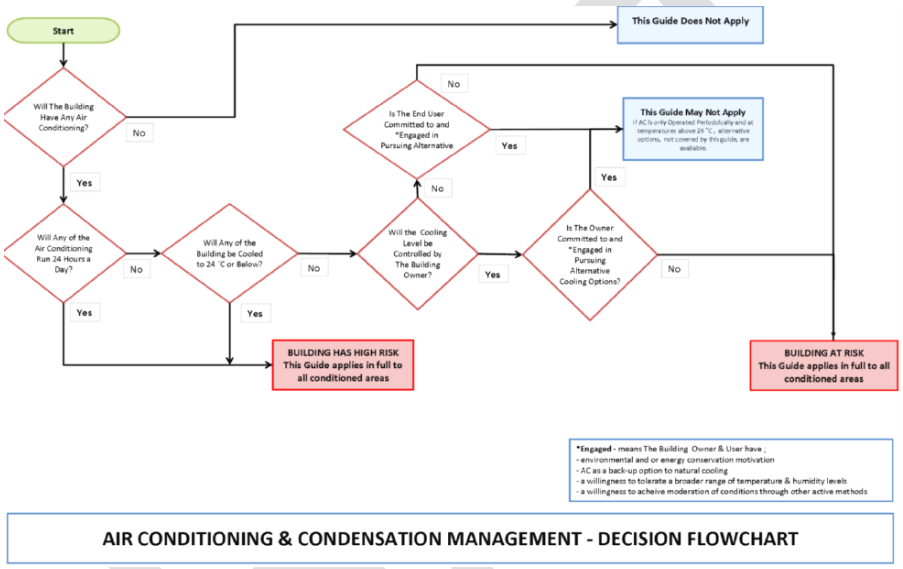

Figure 10 Flowchart of determining air conditioning strategy for building space

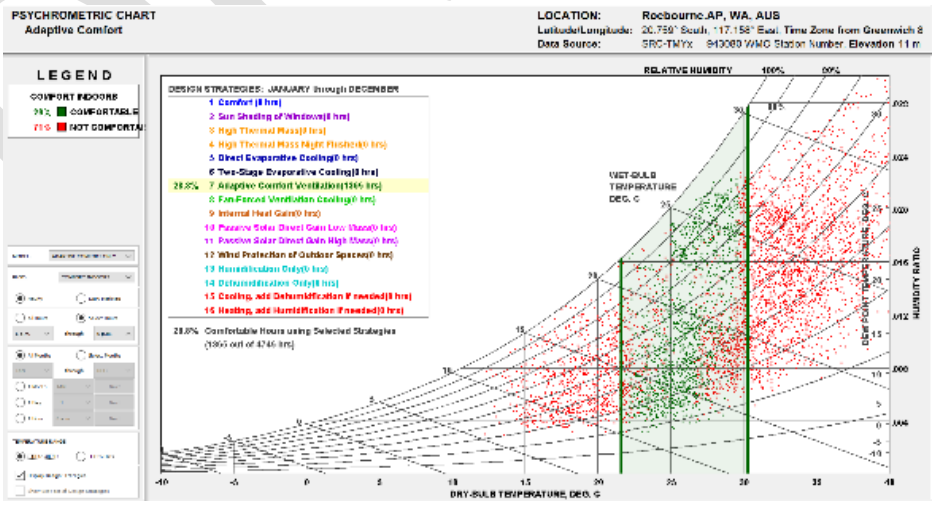

Figure 11 Analysis of potential cooling strategies using psychrometric chart

Notable Example:

- A healthcare campus in Hedland has adopted a courtyard-centric layout. However, the northern building block obstructs prevailing winds, reducing airflow and negatively affecting thermal comfort within the courtyard – an area intended for recreational use.
- To address this, the design team has explored the use of a mixed-mode ventilation system for the reception hall located in the north block. Under natural ventilation mode, increasing the porosity of this space can significantly enhance wind flow into the courtyard. This adjustment would improve the microclimate, boosting thermal comfort and resilience during heatwaves (Figure 12).

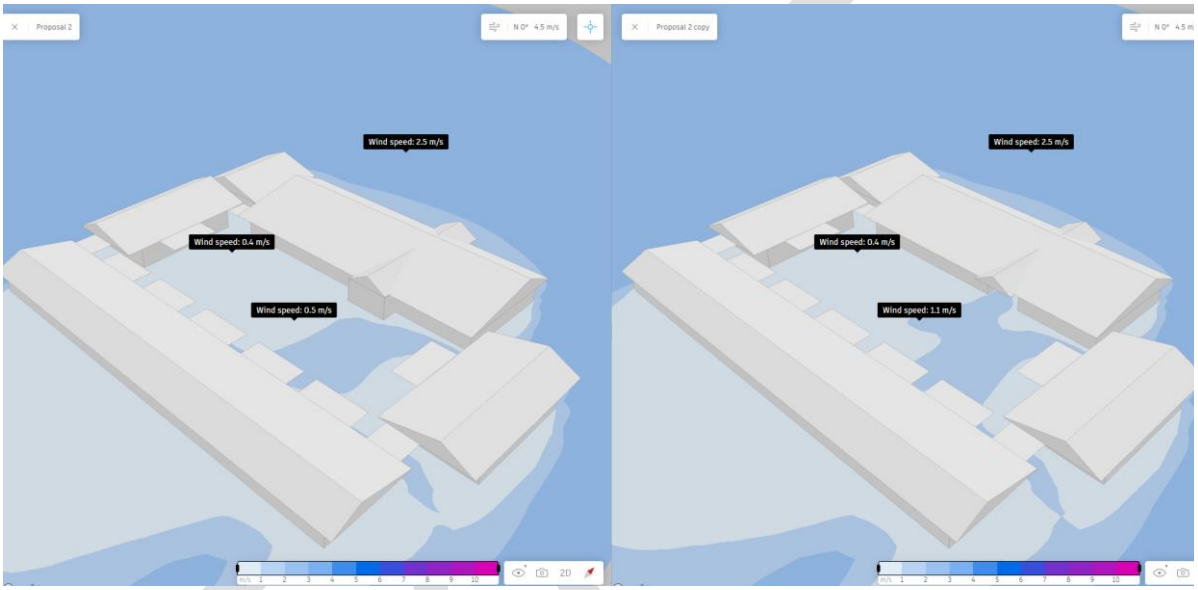

Figure 12 Comparison of building massing options for a healthcare facility

(Left) The buildings are fully air-conditioned; (right) Creation of an opening at the north block allows wind penetration and improves the thermal comfort of the courtyard (the areas where wind speed exceeds 1 m/s is increased by 40%).

4.4. Limiting external heat gain[3]

Once site planning and overall cooling strategies are established, it is essential to begin individual building design by minimising excessive external heat gain through early integration of passive design measures. These include building-integrated solar shading, high-performance insulation in walls and roofs, and glazing systems with optimised window-to-wall ratios and thermally efficient glass.

[3] This section references Clause H3 in Appendix A of the TG050: Climate Change Adaptation Guideline for Non-Residential Government Buildings.

Reducing external heat gain not only enhances occupant thermal comfort but also decreases dependence on mechanical cooling systems, contributing to improved energy efficiency and long-term climate resilience. This section refers to TG050: Appendix A Clause H3.

Recommended approach:

- Apply the passive design measures outlined in Appendix B, tailored to the specific requirements of each climate zone.
- Use thermal modelling to demonstrate a minimum 10% reduction in annual external heat gain – including solar radiation, conductive transfer, and air infiltration – compared to the NCC J1V3 reference building.
- Model simulations should be based on future climate data, specifically high-emission scenarios such as RCP8.5 / SSP3-7.0 for the years 2050 and 2070, to ensure long-term performance under projected climate conditions.

Notable Example:

- Located in Climate Zone 5, a primary school project has implemented a range of passive design strategies to reduce heat gain and improve indoor thermal comfort. Key features include low-emissivity (low-E) glazing with integrated shading, high-performance insulation, infiltration control, and thermally reflective roof coatings (see Figure 13).
- Thermal modelling has confirmed that the design exceeds the energy efficiency requirements of NCC Section J, achieving a reduction in peak cooling loads during summer months by more than 20% (see Figure 14). This demonstrates the effectiveness of passive measures in enhancing comfort and reducing energy demand under current and future climate conditions.

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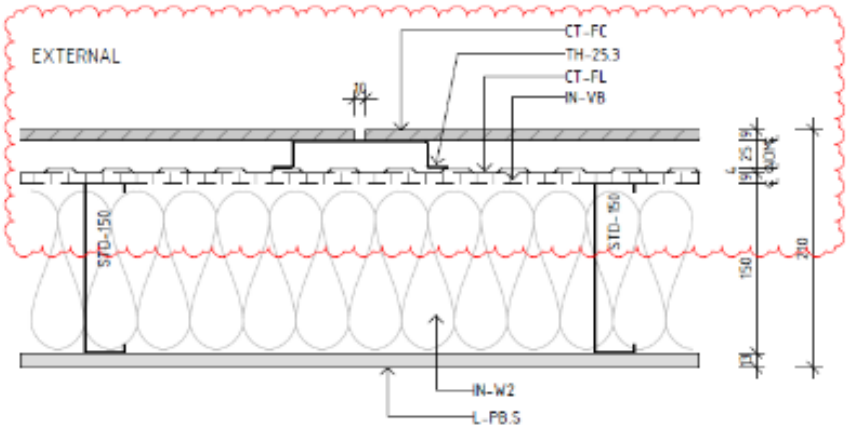
Envelope Feature		Performance		Comment
		BCA Minimum	Documented	
Roofs		R3.7	R4.3	~50mm additional insulation
Roof Lights	U Value	3.9	3.0	PSB achieve these performance levels.
	SHGC	0.51	0.25	
Masonry Walls		R1.4	R1.8	Avoid thermal bridging
Steel Clad Walls		R1.4	R2.3	Avoid thermal bridging
Lightweight Walls		R1.4	R1.4	Avoid thermal bridging
Glass – All buildings, excluding TB2 & TB3	U Value	7.1	4.2	Typical low-e clear glazing
	SHGC	0.80	0.6	
Glass – TB2 & TB3	U Value	7.1	4.2	Typical low-e clear glazing
	SHGC	0.6	0.6	
Floors on Grade		R2.0	>R2.0	Covered Assembly music room, canteen and office will require performance-based solution for compliance.
Exposed Floors		R2.0	R2.2	TB2 & TB3 meeting room floors



*LINING TYPICALLY L-PB.5 UNLESS OTHERWISE NOTED.
REFER TO WALL SETOUT DRAWING FOR ALTERNATE TYPES AND LOCATIONS.

Figure 13 Strategies to limit external heat gains in a primary school in Climate Zone 5

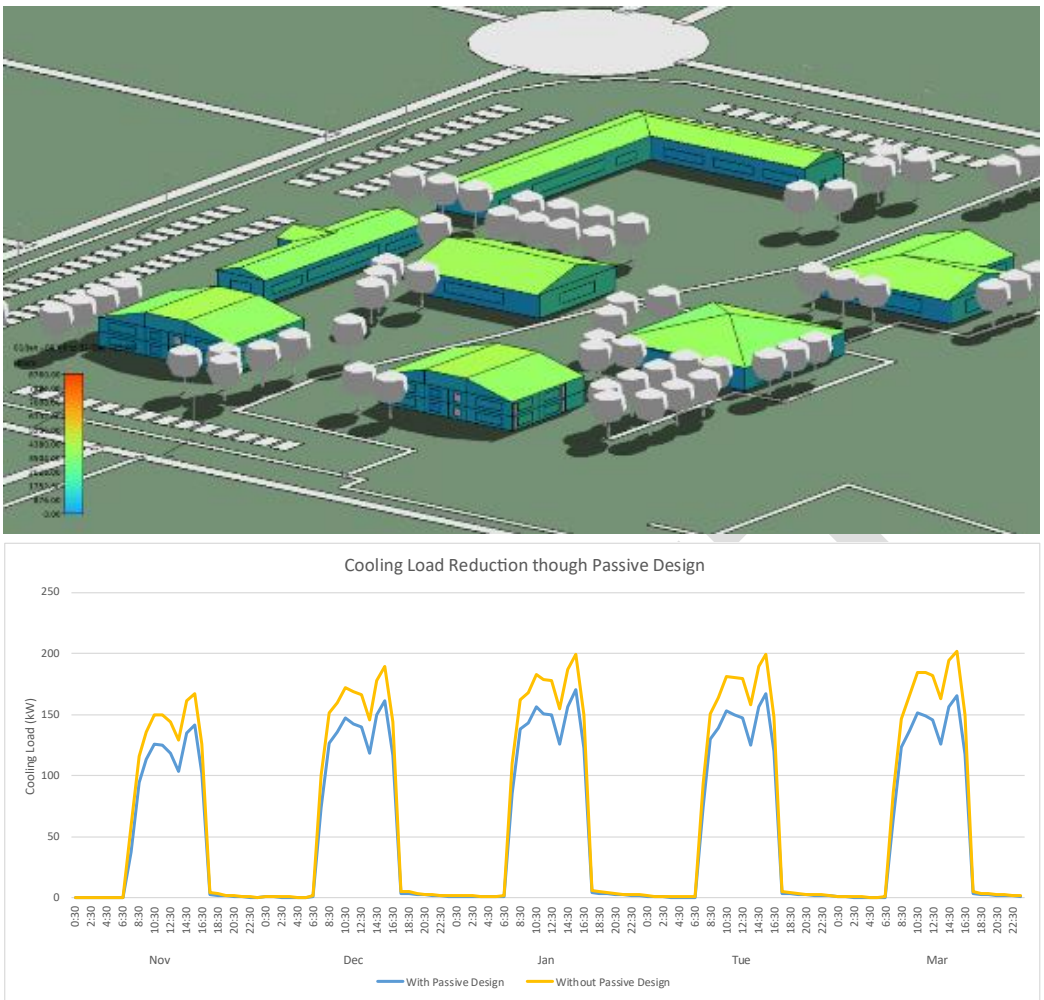


Figure 14 Cooling load reduction in a primary school in Climate Zone 5

4.5. Limiting internal heat gain[4]

After addressing external heat gain, it is equally important to minimise internal heat accumulation within building spaces. This can be achieved by reducing lighting power density, managing plug loads, and implementing operational controls. These measures not only improve thermal comfort but also reduce cooling demand and energy consumption.

<u>Recommended approach:</u>

- Design teams should aim to demonstrate at least a 10% reduction in annual internal heat gain compared to the NCC J1V3 reference building, using thermal modelling and energy performance analysis.

[4] This section references Clause H4 in Appendix A of the TG050: Climate Change Adaptation Guideline for Non-Residential Government Buildings.

Strategies to achieve this include:

- **High-efficiency lighting systems**, such as LED fixtures with high luminous efficacy, combined with smart controls like occupancy sensors and dimmable lighting to reduce unnecessary energy use.
- **Smart plug load management**, including scheduled shut-off settings and programmable timers to automatically disconnect non-essential equipment when not in use.
- **Energy-efficient appliances and equipment**, such as IT devices and audiovisual systems, which reduce both energy consumption and associated heat emissions.

Notable Example

- An assessment of internal heat gain in a 96 m² classroom has established a baseline profile, as illustrated in Figure 15.

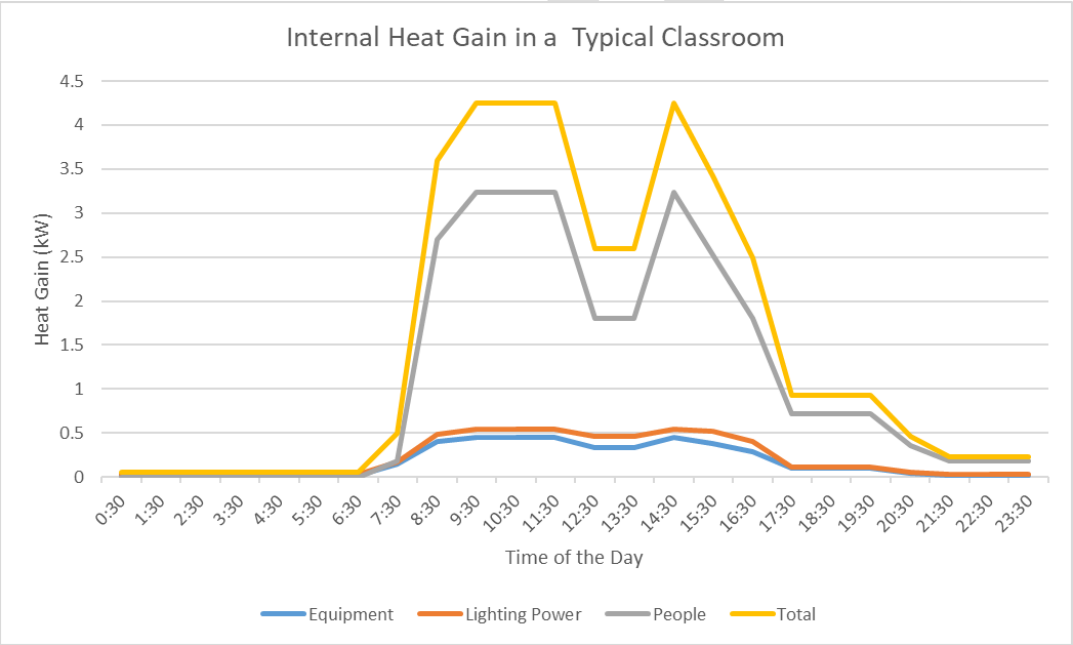


Figure 15 Internal Heat Gain of a Classroom

4.6. Ventilative cooling and heat removal[5]

To maintain acceptable thermal comfort during heatwave conditions, it is essential to reduce reliance on mechanical cooling as the sole strategy –

[5] This section references Clause H5 in Appendix A of the TG050: Climate Change Adaptation Guideline for Non-Residential Government Buildings.

particularly due to the increased risk of power system stress and outages. For critical spaces, alternative ventilation methods should be considered to support demand response measures such as power shedding, ensuring continued functionality and avoiding operational disruptions. In such scenarios, standby power can be prioritised to maintain basic thermal comfort in essential areas.

Recommended strategies include:

- **Hybrid Cooling:** Use ceiling or personal fans either as a supplement or alternative to mechanical cooling. Research shows that increasing air movement can create a ventilative cooling effect, allowing for higher indoor temperatures while still maintaining thermal comfort.
- **Mixed-Mode Ventilation:** In suitable climates – such as northern regions where night-time cooling is available – designs should incorporate multiple ventilation modes, including operable windows, ceiling fans, and air-conditioning. This flexibility allows occupants to select the most appropriate system based on ambient conditions. Low-energy ventilation options also ensure buildings remain functional during power outages[6].
- **Heat Removal:** As hot air rises and accumulates in upper indoor spaces, it is effective to design heat exhaust openings that facilitate stack-effect ventilation. Alternatively, extraction fans can be used to remove warm air from high ceilings, preventing heat buildup and reducing cooling demand.

Notable Examples

- Hybrid cooling systems have been extensively studied in tropical environments, demonstrating multiple benefits including improved thermal comfort and significant energy savings. Recent research indicates that increasing air-conditioning setpoints – from 24°C to approximately 26–27°C – can maintain occupant comfort when combined with enhanced air movement (see Figure 16) (Kent et al., 2023). This adjustment reduces cooling energy demand and helps mitigate the risk of power outages during peak load periods.

[6] Refer to CIBSE AM10: Natural Ventilation in Non-Domestic Buildings (2005) and AM13: Mixed Mode Ventilation Systems (2000).

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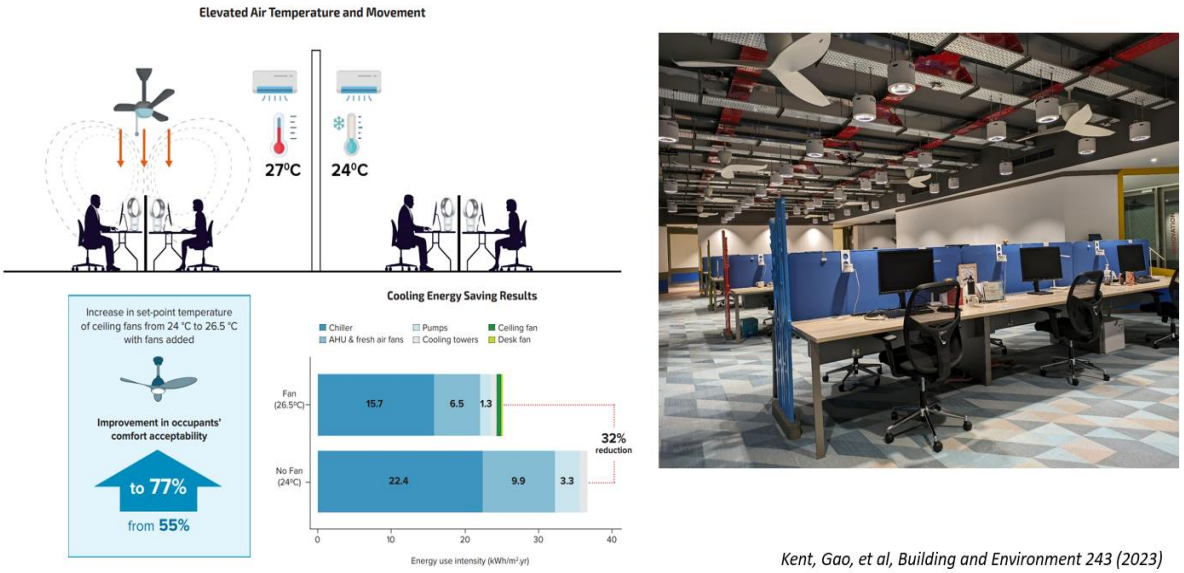

Figure 16 Hybrid cooling in an office environment in Singapore

- In Western Australia, ceiling fans are commonly used in primary and secondary schools, but their application is often limited to classroom settings (see Figure 17). There is significant potential to extend their use to other areas such as administrative offices, libraries, and reception spaces, where they can contribute to improved thermal comfort and reduced reliance on mechanical cooling.

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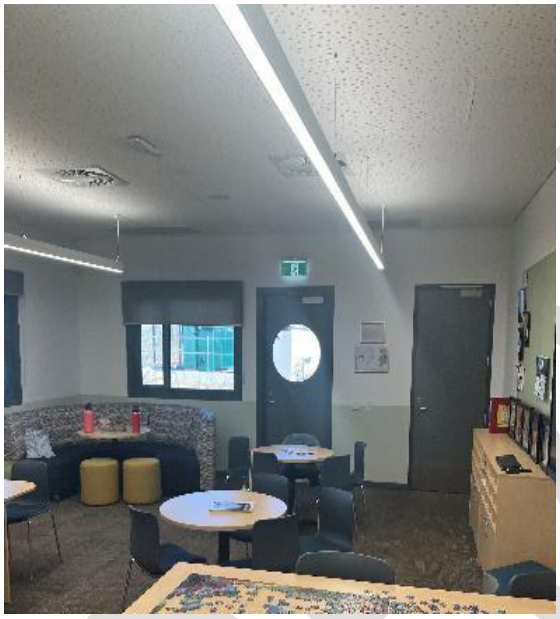
(b) High volume low speed (HVLS) at sport halls

(c) Ceiling fans at classroom



(c) Staff areas are not equipped with ceiling fans

(d) Ceiling fans used in learning spaces

Figure 17 Current use of fans in WA public schools

4.7. Mechanical cooling[7]

Mechanical systems should be designed to accommodate both current and projected thermal loads, particularly in light of the increasing frequency and intensity of heatwaves that pose risks of operational disruption and downtime. Cooling systems must be appropriately sized and configured with operating parameters that ensure reliable performance under extreme heat conditions.

[7] This section references Clause H6 in Appendix A of the TG050: Climate Change Adaptation Guideline for Non-Residential Government Buildings.

Given the limited service life of mechanical systems, it is essential to incorporate design provisions that facilitate future upgrades or replacements. This includes ensuring adequate spatial allowances for equipment expansion or retrofitting.

To support this process, a checklist outlining key considerations for designing mechanical cooling systems to meet future thermal demands is provided in **Appendix C**.

Key recommendations include:

- **Design for Heatwave Resilience:** HVAC systems should be sized and specified to operate effectively under heatwave conditions, reducing the risk of system failure and operational disruption.
- **Use Updated Climate Data:** Define HVAC design conditions using the most current climate datasets available from sources such as the *ASHRAE Handbook—Fundamentals* and *AIRAH DA09: Air Conditioning Load Estimation and Psychrometrics*. Project teams should also incorporate site-specific data from the Bureau of Meteorology (BoM) to account for the increasing frequency of extreme weather events.
- **Incorporate Climate Change Projections:** Assess future increases in temperature and humidity using climate projection datasets from *Climate Change in Australia (CCiA)* and *CSIRO*, particularly under the RCP8.5 / SSP3-7.0 scenarios for 2050 and 2070. Use these insights to adjust design conditions and ensure appropriate equipment sizing and specifications.
- **Plan for Future Upgrades:** Allocate additional plant space to accommodate larger or more efficient thermal equipment, allowing flexibility for future retrofits and system enhancements.
- **Optimise Equipment Placement:** Relocate outdoor HVAC units to shaded and well-ventilated areas to improve operational efficiency and reduce thermal stress.
- **Balance Cooling and Heating Needs:** While cooling demand is expected to rise, there may be opportunities to downsize heating equipment due to milder winters associated with climate change.

Notable Example:

- A design review of an educational building in Roebourne revealed that the HVAC engineers had based their design on outdated climate data from 1975–1984 (see Table 2). As a result, the design temperature was approximately 1°C lower than the prevailing standard derived from more

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recent data (2000–2014), leading to an underestimation of cooling requirements.

- Further analysis using a GIS-based climate mapping tool – developed with high-resolution data from CSIRO (5 km resolution) – projected a temperature increase of up to 1.71°C in Roebourne between 2000 and 2050 under the RCP8.5 scenario (see Figure 18). Additionally, the number of extreme hot days exceeding 35°C is expected to rise from 169 to 201 days. Given that the HVAC equipment is scheduled for replacement around 2040, the current design is unlikely to meet future thermal demands.
- In light of these findings, it is recommended that design consultants incorporate climate change projections into their specifications and adjust design parameters accordingly. This should be reflected in both tender documentation and contractor selection to ensure long-term resilience and performance of mechanical systems.

Table 2 Design Conditions for a Project in Roebourne, WA

Site	AIRAH design condition (1975-1984 3pm) (A)	AIRAH design condition (1990-2013 3pm) (B)	AIRAH design condition (2000-2014 3pm) (C)	ASHRAE design condition (2021 1% monthly) (D)	Project A's design condition (E)	Buffer (F = E – C)	Projected temperature rise by 2050 (RCP8.5) (G)	Is the buffer sufficient for climate change (F > G)?
Roebourne	42.6ºC (DB) 27.8ºC (WB)	43.0ºC (DB) 29.6ºC (WB)	43.6ºC (DB) 31ºC (WB)	43.43ºC (DB) 24.74(WB)	42.6ºC (DB) 27.8ºC (WB)	-1.0ºC	1.71ºC	No

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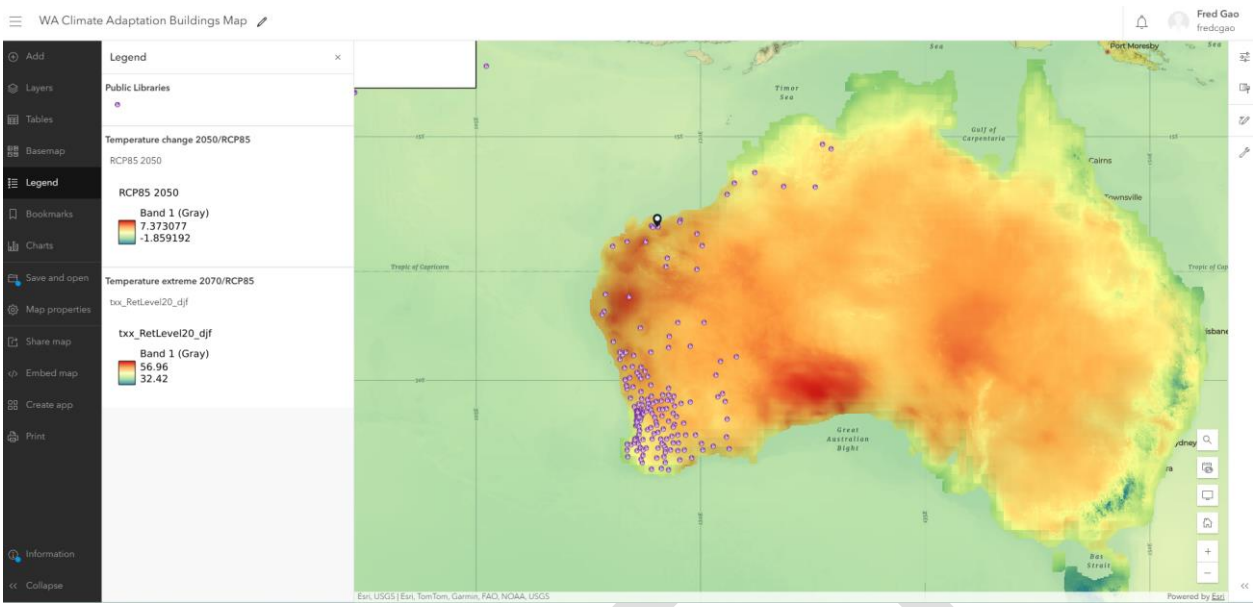

Figure 18 GIS Tool for Determining Design Conditions for Climate Adaptation

4.8. Heat resilient materials[8]

To ensure long-term thermal performance and durability under extreme heat conditions, it is essential that designers specify building materials capable of withstanding elevated temperatures. Key strategies include:

- **Integrating Thermal Mass**: Incorporate thermal mass into the building envelope to moderate indoor temperatures and reduce peak cooling demand. A target reduction of at least 2% compared to a reference NCC-compliant building is recommended.
- **Selecting Thermally Stable Materials**: Use materials that maintain structural integrity and performance under high temperatures. These should resist deformation, degradation, and loss of function under projected climate conditions.
- **Specifying Low-Conductivity Materials**: For elements such as handrails and outdoor fixtures, select materials with low thermal conductivity – such as timber or recycled plastic – to minimize heat transfer and reduce the risk of occupant discomfort or injury (refer to ISO 13732-1:2006).

[8] This section references Clause H7 in Appendix A of the TG050: Climate Change Adaptation Guideline for Non-Residential Government Buildings.

- **Choosing UV-Stable and Heat-Durable Finishes**: Roofs, walls, and floors should be constructed with materials that are resistant to UV radiation and capable of enduring prolonged exposure to extreme temperatures.

Notable Example

- Surface coatings and reflective materials are effective strategies for enhancing heat resilience in buildings and surrounding environments. When applied to roofs, walls, and outdoor pavements, these treatments can significantly reduce ambient air temperatures and lower energy consumption within buildings.
- Advanced technologies – such as nano-engineered coatings that incorporate microscopic reflective elements – can further improve thermal performance by reflecting solar radiation and minimizing heat absorption.
- A comparative study has shown that cool paints and pavements can reduce surface temperatures by up to 20°C on extremely hot days. This translates to a reduction in ambient air temperature of approximately 1–2°C, contributing to a more comfortable thermal environment both inside and outside the building (see Figure 19).

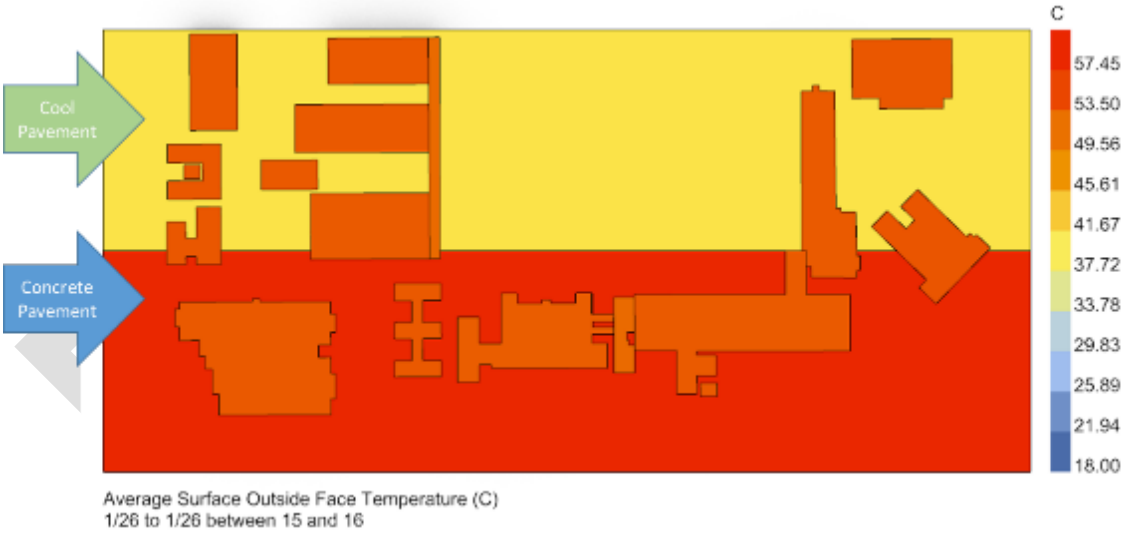

Figure 19 Surface temperature comparison between heat reflective pavement and concrete pavement

Appendix A: Rising Temperature in WA

Climate change is intensifying summer heat across Australia, with Western Australia experiencing some of the most pronounced effects. The decade of 2015 - 2024 was the warmest on record since thermometer-based observations began, with 2023 and 2024 the hottest years in a row. Heatwaves – defined as three or more consecutive days of unusually high daytime and night-time temperatures – are becoming more frequent, longer, and more severe (Figure 20) (CSIRO and BOM, 2024).

Since the 1930s, the number of days exceeding 40°C in Perth has doubled, and heatwave occurrences have increased by 50%. Between 1910 and 2024, the state's average temperature has risen by approximately 1.5°C.

Extended periods of high temperatures pose serious health risks, particularly due to the strain they place on the cardiovascular system. This risk is heightened when night-time temperatures remain elevated, preventing the body from recovering. The degree of exposure to heat during these events is significantly influenced by the thermal performance of buildings – especially those without adequate cooling or passive design features.

Extreme heat is Australia's deadliest natural hazard. As shown in Figure 21, heatwaves have accounted for approximately 55% of all natural hazard-related fatalities recorded in the PerilAUS database since 1900 – more than any other hazard, and more than all other hazards combined (Perkins-Kirkpatrick et al., 2016). Furthermore, a recent study by the Department of Health projects that heat-related deaths in Perth could rise by 61.4% in the 2050s compared to the 2010s (Sharma S, 2024).

Heat also affects productivity. As temperatures climb, fatigue increases, concentration declines, and the likelihood of accidents rises. The threshold at which heat begins to impair work capacity varies depending on acclimatisation and the nature of the work. This has important implications for workers in buildings without air conditioning or adequate thermal comfort measures.

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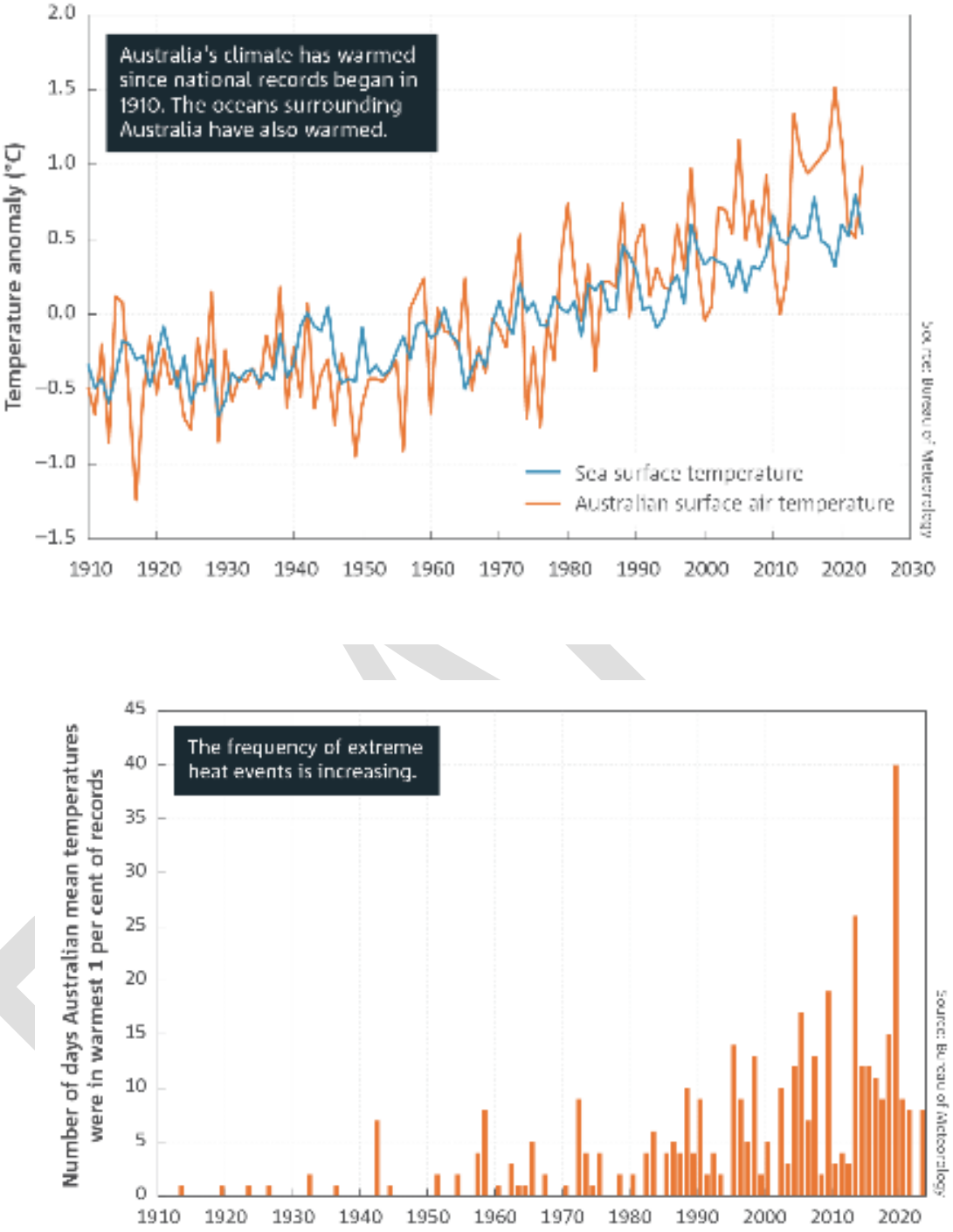

Figure 20 Australia's temperature rise and increase of extreme heat events

(Top) Australia's climate has warmed by an average of 1.51 ± 0.23 °C since national records began in 1910; (Bottom) There has been an increase in extreme heat events associated with the warming. (Source: CSIRO, BOM)

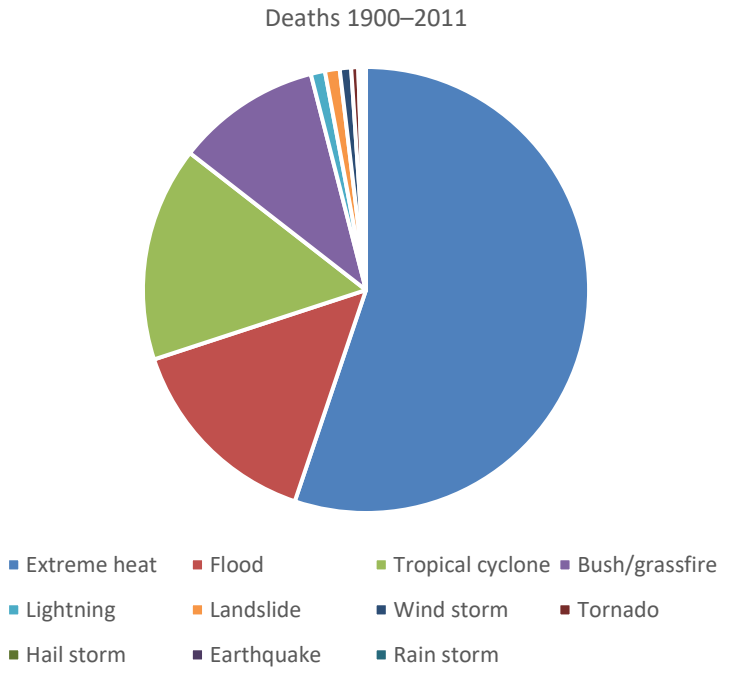

Figure 21 Fatality totals of Australian natural hazards from 1900 - 2011

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Appendix B: Climate Zone Specific Design Solutions

General

The majority of government non-residential buildings in Western Australia are air-conditioned, and with increasing temperatures, air-conditioning use is also likely to increase (with associated increases in greenhouse gas emissions). As most of these buildings are occupied during the day, and most people are often unable to change their location in a building, control of over-heating in spaces is very important.

General measures to prevent or minimise building over-heating in non-residential government buildings include:

- Use of reflective glazing and external shading (trade- off between preventing heat entry and natural light penetration).
- Increasing insulation and use of thermal mass. Inclusion of passive ventilation methods.
- Use of integrated automation controls, to integrate lighting, HVAC, ventilation and security systems.
- Reducing lighting and equipment loads.
- Use of energy-efficient lighting and equipment (less heat produced).
- Use of light sensors which will automatically dim levels to utilise natural light.
- Upgrading the air-conditioning system to maximise operational efficiency.

An overview of actions for each climate area is provided below (responsibility for the implementation of these actions is spread across planning and development regulations, building regulations, and personally through lifestyle choice)

Zone 1 (Hot Humid): Broome

- Employ lightweight (low mass) construction.
- Maximise external wall areas (plans with one room depth are ideal) to encourage movement of breezes through the building (cross-ventilation).
- Site new buildings for exposure to breezes and shading all year.

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- Shade whole building summer and winter (consider using a fly roof).
- Use reflective insulation and vapour barriers.
- Ventilate roof spaces.
- Use bulk insulation if mechanically cooling.
- Choose light-coloured roof and wall materials.
- Elevate buildings to permit airflow beneath floors.
- Consider high or raked ceilings.
- Provide screened, shaded outdoor areas.
- Design and build for cyclonic conditions.

Zone 3 (Hot Dry, Warm Winter): Carnarvon

- Use passive solar design with insulated thermal mass
- Maximise cross-ventilation.
- Use air purging to remove the accumulated heat
- Consider convective (stack) ventilation, which vents rising hot air while drawing in cooler air.
- Site new buildings for solar access and exposure to cooling breezes.
- Shade all east and west glass in summer.
- Install reflective insulation to keep out heat in summer.
- Use bulk insulation in ceilings and walls.
- Build screened, shaded summer outdoor areas that allow winter sun penetration.

Zones 4 and 6 (Hot Dry, Cool/Cold Winter): Geraldton, Albany

- Use passive solar principles with well-insulated thermal mass.
- Maximise night-time cooling in summer.
- Consider convective (stack) ventilation, which vents rising hot air while drawing in cooler air.
- Use air purging to remove the accumulated heat
- Build more compact shaped buildings with good cross-ventilation for summer.
- Maximise solar access, exposure to cooling breezes or cool air drainage, and protection from strong winter (cold) and summer (dusty) winds.
- Shade all east and west glass in summer.
- Provide shaded outdoor areas.
- Consider adjustable shading to control solar access.
- Auxiliary heating may be required in extreme climates. Use renewable energy sources.
- Use evaporative cooling if required.
- Avoid long operation hours of air-conditioning.
- Use reflective insulation for effective summer and winter application.
- Use bulk insulation for ceilings, walls and exposed floors.
- Draught seal thoroughly. Use airlocks to entries.

Zone 5 (Temperate): Perth

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- Use passive solar principles.
- High thermal mass solutions are recommended.
- Use high insulation levels, especially to thermal mass
- Use air purging to remove the accumulated heat
- Maximise north facing walls and glazing, especially in living areas with passive solar access.
- Minimise all east and west glazing. Use adjustable shading.
- Use heavy drapes with sealed pelmets to insulate windows.
- Minimise external wall areas (especially E & W).
- Use cross-ventilation and passive cooling in summer.
- Encourage convective ventilation and heat circulation.
- Site new buildings for solar access, exposure to cooling breezes and protection from cold winds.
- Draught seal thoroughly and use entry airlocks.
- No auxiliary heating or cooling is required in these climates with good design.
- Use reflective insulation to keep out summer heat.
- Use bulk insulation to keep heat in during winter.
- Bulk insulate walls, ceilings and exposed floors.

[Note: Although Zone 5 is generally considered to have a more preferable thermal environment than other zones, buildings in Zone 5 are particularly vulnerable to climate change as the building design and construction practice there typically do not place emphasis on thermal resistance to extreme outdoor environment.]

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Appendix C: Mechanical Systems Design Checklist

1. Climate-Responsive Design Inputs

- ☐ Use latest climate datasets (e.g. ASHRAE Handbook, AIRAH DA09) for HVAC design conditions.
- ☐ Incorporate site-specific data from BoM for extreme weather frequency.
- ☐ Apply future climate projections (RCP8.5 / SSP3-7.0 for 2050 & 2070) from CSIRO or CCiA.

2. Equipment Sizing and Specification

- ☐ Size HVAC systems to operate effectively under heatwave conditions.
- ☐ Adjust design setpoints and operating parameters based on projected temperature and humidity increases.
- ☐ Consider downsizing heating systems due to reduced winter demand.

3. Flexibility and Future-Proofing

- ☐ Allocate additional plant space for future equipment upgrades.
- ☐ Ensure access and layout allow for easy replacement or expansion of mechanical systems.
- ☐ Design modular systems where possible to support phased upgrades.

4. Equipment Placement and Efficiency

- ☐ Locate outdoor HVAC units in shaded, well-ventilated areas.
- ☐ Avoid placing equipment in heat-prone zones or areas with poor airflow.

5. Resilience and Redundancy

- ☐ Include provisions for standby power to maintain basic thermal comfort in critical spaces.
- ☐ Design systems to support demand response strategies (e.g. power shedding).
- ☐ Integrate low-energy ventilation options (e.g. ceiling fans, operable windows) for passive cooling during outages.

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Appendix D: Heat Resilience Strategies

The table below is sourced from Appendix A of the TG050: Climate Change Adaptation Guideline for Non-Residential Government Buildings.

Table A1.1 - Heat
H1. Thermal Comfort Design Criteria (Rising Temperature) Thermal Comfort (TC) design targets and evaluation criteria should be defined early in the design process, taking into account projected climate change impacts and the specific vulnerabilities of building occupants. The following criteria are recommended for consideration. For mechanically ventilated spaces: ○ Achieve a Predicted Mean Vote (PMV) between -1.0 and +1.0 for not less than 95% of the occupied floor area for not less than 98% of the building's operational hours, based on suitable future weather files. ○ For spaces for vulnerable occupant groups, such as the elderly, disabled, infirm or very young children, achieve a PMV between 0 and 1.0 for 95% of the occupied floor area for 98% of the building's operational hours, based on suitable future weather files. For building spaces that rely on natural ventilation or mixed mode ventilation systems: ○ Achieve acceptable operative temperatures with at least 80% acceptability across 95% of the occupied floor area for 98% of the building's operational hours following the adaptive thermal comfort model**, based on suitable future weather files. ○ For spaces for vulnerable occupant groups, such as the elderly, disabled, infirm or very young children, achieve acceptable operative temperatures with at least 90% acceptability across 95% of the occupied floor area for 98% of the building's operational hours following the adaptive thermal comfort mode, based on suitable future weather files. Notes: ○ Refer to NCC 2022 NCC J1V3(1)(b), NCC JP1 Guide on Adaptive Thermal Comfort and PMV, and ASHRAE Standard 55-2020 Thermal Environmental Conditions for Human Occupancy. ○ Future weather files incorporate projected climate change data to enable dynamic simulations of building performance under future conditions. Project teams are encouraged to use the SSP3-7.0 scenario for the years 2050 or 2070 (or an equivalent projection based on RCP8.5) when conducting these assessments. ○ Refer to ISO 28803:2012 Ergonomics of the Physical Environment – Application of International Standards to People with Special Requirements for thermal comfort requirements for vulnerable populations.

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H2. Thermal Comfort Design Criteria (Heatwave)

Buildings should be designed to maintain thermal comfort for occupants during heatwaves. It is acknowledged that maintaining thermal comfort in all spaces, at all times, during heatwaves may not be possible. Therefore, priority should be given to maintaining thermal comfort in critical areas, such as those with vulnerable occupants or essential functions. To achieve this, the following control measures are recommended:

For critical spaces serving vulnerable user groups in Importance Level (IL) 3 or IL 4 buildings – such as aged care homes or patient wards – indoor thermal conditions should comply with the CIBSE TM52 Category I standard, using future weather files for simulation.:

- In mechanically cooled spaces, the number of overheating hours – defined as periods exceeding the PMV/PPD or operative temperature thresholds – should not exceed 3% of total occupancy hours.
- In naturally ventilated or mixed mode ventilation spaces, overheating hours – defined as periods exceeding the TM52 criteria – should not exceed 3% of total occupancy hours.

Note:

- Refer to TM52-2013 The Limits of Thermal Comfort: Avoiding Overheating, Section 6.1 and 6.2.
- The future weather data reflecting extreme heat conditions impacted by climate change include:
    - Future Extreme Meteorological Year (fXMY),
    - Future Design Summer Year (fDSY), and

  In the absence of the above, CSIRO's projected Typical Meteorological Year (TMY) files based on the RCP8.5/ SSP3-7.0, 2090 scenario may be used as a proxy.

H3. Limit External Heat Gain through Passive Design

Incorporate passive design strategies to minimise excessive external heat gain accumulation in buildings. This can enhance thermal comfort and reduce reliance on mechanical cooling systems. The following approach is recommended:

- Use thermal modelling to demonstrate at least a 10% reduction in annual external heat gain (from solar radiation, conductivity and infiltration) compared to the NCC J1V3 reference building.
    - Modelling should be based on climate data reflecting future high-emission scenarios, specifically the RCP8.5 / SSP3-7.0, 2050 and 2070 scenario.

H4. Limit Internal Heat

Internal heat accumulation can be minimised by reducing lighting power and managing plug loads. To achieve this, the following control measures are recommended:

- Through thermal modelling, demonstrate at least a 10% reduction of annual internal heat gain as compared to the NCC J1V3 reference building using climate data based on RCP8.5/ SSP3-7.0, 2050 and 2070 scenario. A combination of energy efficiency and control measures is recommended, including but not limited to:

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- High efficiency light fittings and controls, such as use of high efficacy LED lighting, occupancy sensors, dimmable lighting systems.
- Smart plug load controls, such as scheduled shutoff settings and programmable timers should be installed to automatically disconnect non-essential equipment when not in use.

H5. Mechanical Systems and Control Strategies

Mechanical systems should be designed to meet future thermal demands. To achieve this, the following control measures are recommended:

- HVAC design conditions should be defined using the most up-to-date climate data available from sources such as the ASHRAE Handbook – Fundamentals or AIRAH DA09: Air Conditioning Load Estimation and Psychrometrics. Additionally, project teams should consider the likelihood of recurring extreme weather events, using the latest project-specific data from the Bureau of Meteorology (BoM).
- Assess potential increases in temperature and humidity using the latest climate change projection data, including datasets from Climate Change in Australia (CCiA) and CSIRO's projections under the RCP 8.5 / SSP3-7.0 scenarios for 2050 and 2070. Use these insights to adjust design conditions, accordingly, ensuring appropriate equipment sizing and technical specifications.
- Allow for additional plant and riser space in the design to accommodate larger thermal equipment, ensuring flexibility for future upgrades and climate resilience.
- Relocate HVAC outdoor equipment into optimal spaces with improved shading and ventilation.
- Implement active control strategies through Building Management Systems (BMS) and/or timer controls. Recommended measures include demand-controlled ventilation, temperature setpoint adjustments, zonal controls, and night-time purging/cooling.

Note: While cooling load is expected to increase, there will be an opportunity of downsizing heating equipment due to climate change.

H6. Natural Ventilation and Air Flow Management

Design effective natural ventilation systems to manage heat and moisture, reduce dependence on mechanical cooling and power systems. Strategies may include:

- Single-sided or cross ventilation using operable windows to promote airflow.
- Stack-effect ventilation to extract internal heat, particularly in spaces with high ceilings.

Note: Refer to CIBSE AM10 Natural ventilation in non-domestic buildings (2005), AM13 Mixed mode ventilation systems (2000).

H7. Heat Resilient Materials

Select building materials that enhance heat resilience and support long-term thermal performance. Key strategies include:

- Incorporating thermal mass into the building fabric to reduce peak cooling demand, with a target of at least a 2% reduction compared to a reference code-compliant building under the NCC.
- Using materials with high thermal stability and durability under elevated temperatures, ensuring they do not deform, degrade, or lose performance under projected climate conditions.

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- Specifying low-conductivity materials – such as timber or recycled plastic – for handrails and outdoor equipment to minimize heat transfer to occupants.
- Select roof, wall and floors materials for increased durability to extreme temperatures (for example, UV-stable materials).

Note: Refer to ISO 13732-1:2006 Ergonomics of the thermal environment – Methods for the assessment of human responses to contact with surfaces. Part 1: Hot surfaces.

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Standards

Standard	Description and relevance
AS 5334:2013 Climate change adaptation for settlements and infrastructure – a risk-based approach	Australian standard that provides a framework and guidelines for assessing risks to infrastructure and settlements from climate change. It provides a structure and approach to planning and adaptation based on the risk management process in ISO 31000 (Standards Australia, 2013).
ISO 14090:2019 Adaptation to climate change: principles, requirements and guidelines	International standard that outlines guidelines and requirements for adapting to climate change. It is a non-linear approach so organisations can implement the principles at different stages. Organisations can use it to inform decisions and to better understand the impacts of climate change (ISO, 2019).
ISO 14091:2021 Adaptation to climate change: guidelines on vulnerability, impacts and risk assessments	International standard that outlines the process for assessing risks related to the impacts of climate change. It provides guidance on understanding vulnerability and developing a risk assessment for the impacts of climate change. Also, while giving understanding of vulnerabilities, it gives structure to preventing or mitigating the impacts while also considering the opportunities (ISO, 2021).
ISO 28803:2012 Ergonomics of the Physical Environment – Application of International Standards to People with Special Requirements	This International Standard describes how International Standards concerned with the ergonomics of the physical environment can be applied for people with special requirements, who would otherwise be considered to be beyond the scope of those standards. It has been produced according to the principles of accessible design provided in ISO/IEC Guide 71 and using the data provided in ISO/TR 22411.
ISO 13732-1:2006 Ergonomics of the thermal environment – Methods for the assessment of human responses to contact with surfaces. Part 1: Hot surfaces.	ISO 15536-1:2006 provides temperature threshold values for burns that occur when human skin is in contact with a hot solid surface. It also describes methods for the assessment of the risks of burning, when humans could or might touch hot surfaces with their unprotected skin.
AS1668.2:2024 The use of ventilation and air-conditioning in buildings, Part 2: Mechanical ventilation in buildings	AS 1668.2 2012 sets out requirements for mechanical air-handling systems that ventilate buildings and car parks, and for ventilation based on the need to control odours, particulates and specific gases. Also known as AS 1668 part 2.

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Useful data sources

- CSIRO, Projections tools <https://www.climatechangeinaustralia.gov.au/en/projections-tools/>
- CSIRO & Bureau of Meteorology, <https://www.climatechangeinaustralia.gov.au>
- Bureau of Meteorology, Climate data online <http://www.bom.gov.au/climate/data/>
- Geoscience Australia, Community safety data and products <https://www.community- safety.ga.gov.au/data-and-products>
- Government of Western Australia, Western Australian climate projections
- Climate Council, Climate risk map of Australia

  <https://www.climatecouncil.org.au/resources/climate-risk-map/>

Useful Resources for acquiring Climate Data

- Bureau of Meteorology: Climate Data Online portal and State of the Climate report
- CSIRO Climate Change in Australia website
- Geosciences Australia specialist data sources
- Climate Council Climate Risk Map of Australia

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